SMART CITY BIKE MOBILITY & STATION ANALYTICS

Project Overview and Objective

Project Overview

This project, titled "Smart City Bike Mobility & Station Analytics," is designed to evaluate the operational performance and infrastructure distribution of urban bike-sharing networks. The analytical framework aggregates data across over 1,000 stations, 57K total bike stands, and 20K available bikes. The data model integrates dimension and fact tables—such as Dim Station (containing station metadata, addresses, and geographic coordinates) and Fact_Bike_Activity (capturing operational status, occupancy rates, and time stamped activity logs)—to enable comprehensive mobility insights.

Objective

Operational Status Analysis: Evaluate the distribution and ratio of open versus closed bike stations across the entire network.

Resource Allocation Assessment: Analyse the relationship between total infrastructure capacity (bike stands) and actual inventory levels (available bikes).

Temporal Trend Tracking: Monitor occupancy rate fluctuations and usage patterns over time (spanning 2022 to 2025).

Geographic & Contract Management: Leverage spatial visual mapping and contract-level slicers to streamline regional service delivery across different municipal networks.

Problem Statement

1. Operational Status Imbalance:

The network currently exhibits an equal 50%-50% split between active and inactive stations (1K Open vs. 1K Closed). Identifying closed stations and uncovering the underlying reasons for their inactivity is critical to meeting commuter demand and restoring service availability.

2. Capacity vs. Inventory Mismatch:

While total stand capacity is high (57K), the overall number of available bikes is significantly lower (20K). Analyzing this gap helps pinpoint localized supply shortages and optimizes bike rebalancing operations across high-demand zones.

3. Underutilization & Maintenance Tracking:

Multiple stations demonstrate low or 0% occupancy rates along with stale update timestamps. Detecting these underperforming stations from real-time activity logs is essential for proactive maintenance and fleet reallocation.

4. Multi-Contract & Regional Disparities:

Service performance varies widely across different contracts and cities (e.g., Lyon, Toulouse). Establishing regional benchmarks and tracking banking availability helps resolve infrastructure bottlenecks in specific municipalities

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Attribute (Column /Features) Details

Attribute Name	Data Type	Description
Number	Integer (Whole Number)	Unique identifier for each bike station (Primary/Foreign Key connecting station metadata with activity logs).
Name	Text / String	Full official name of the bike station (e.g., City Central Bike Station 43).
Address	Text / String	Physical street and city address where the station is located (e.g., 144 Main Street, City).
Position	Text / Geographic (Lat, Long)	Geographical coordinates used for spatial mapping and plotting on visuals.
Contract Name	Text / String	Name of the city or municipal network contract managing the station (e.g., Lyon, Toulouse).
Contractor ID	Text / String	Unique code identifying the specific operating contractor for the station network.
Banking	Boolean (True/False)	Indicates whether the station supports card/digital payment transactions on-site.
Status	Text / String	Operational availability status of the station (OPEN or CLOSED).
Bonus	Boolean (True/False)	Indicates if special rewards or incentive bonus points apply at the station.
Bike Stands	Integer (Whole Number)	Total docking capacity available at the station.
Available Bike Stands	Integer (Whole Number)	Number of empty docks currently open for returning bikes.
Available Bikes	Integer (Whole Number)	Total number of bikes ready for rental at the station.
Last Update	Date/Time	Timestamp indicating when the station data was last refreshed in the system.
Occupancy Rate%	Percentage / Decimal	Calculated metric representing the percentage of total stand capacity currently occupied.

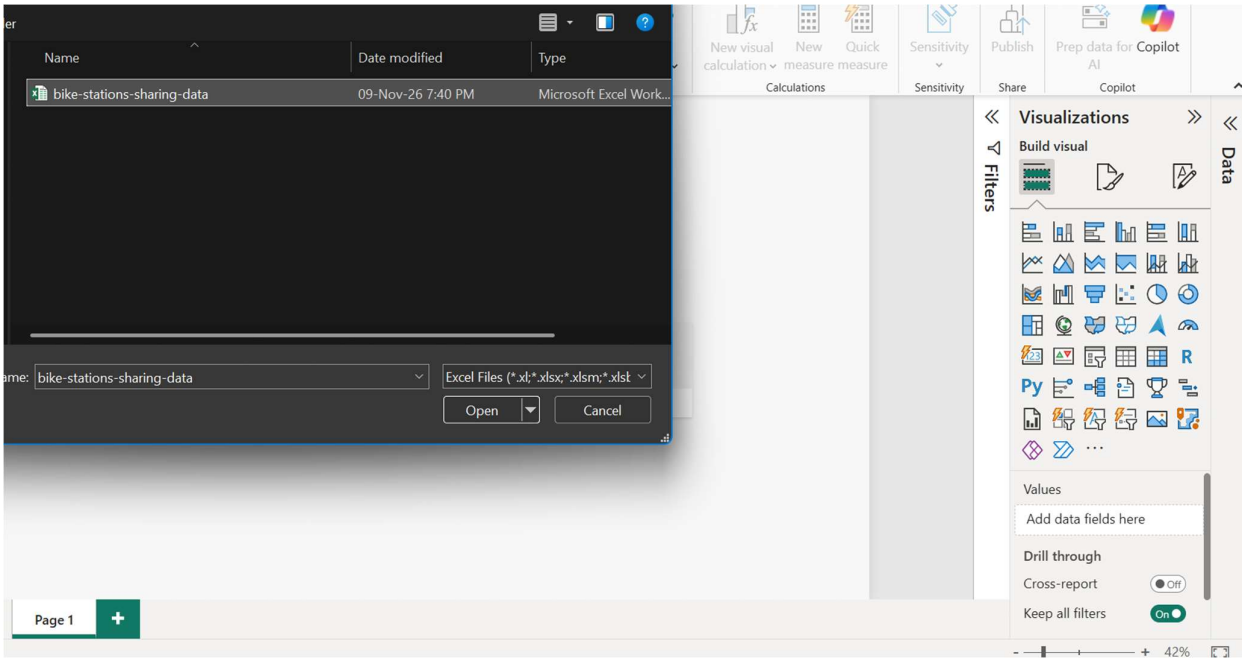
Tools & Technologies

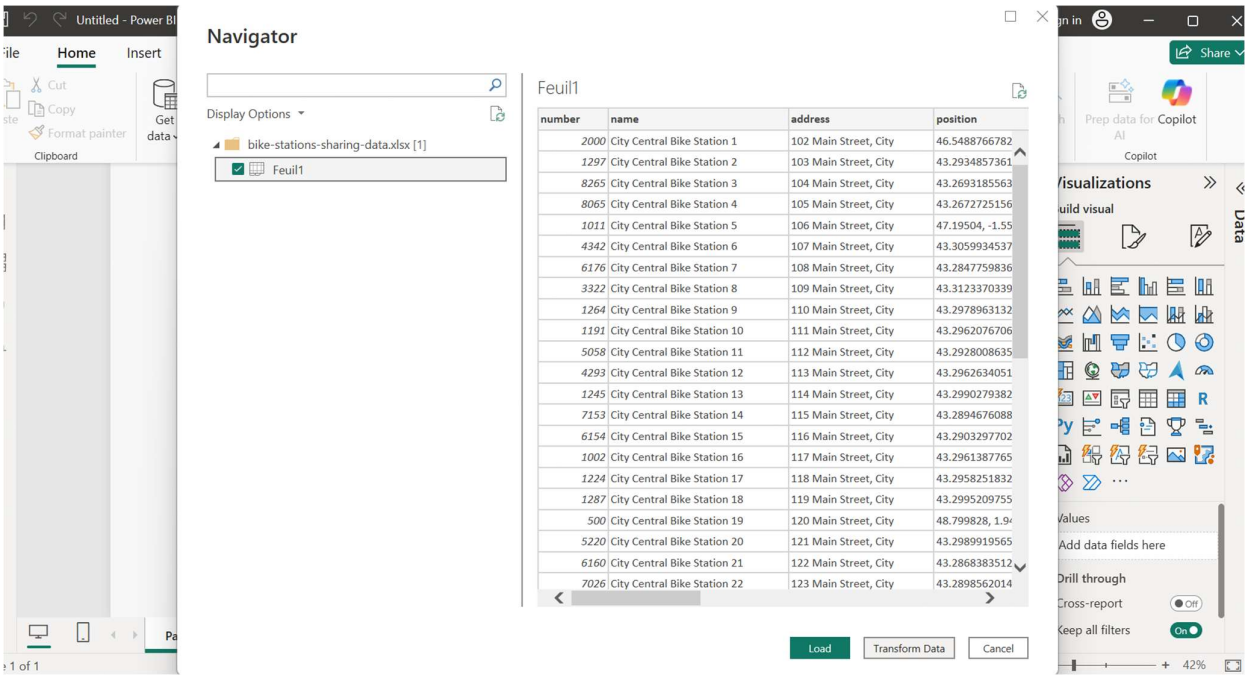
Power BI: Data modelling, DAX calculations, visualization, and interactive dashboard creation.

Data Pre-Processing (Power Query)

Tasks Performed:

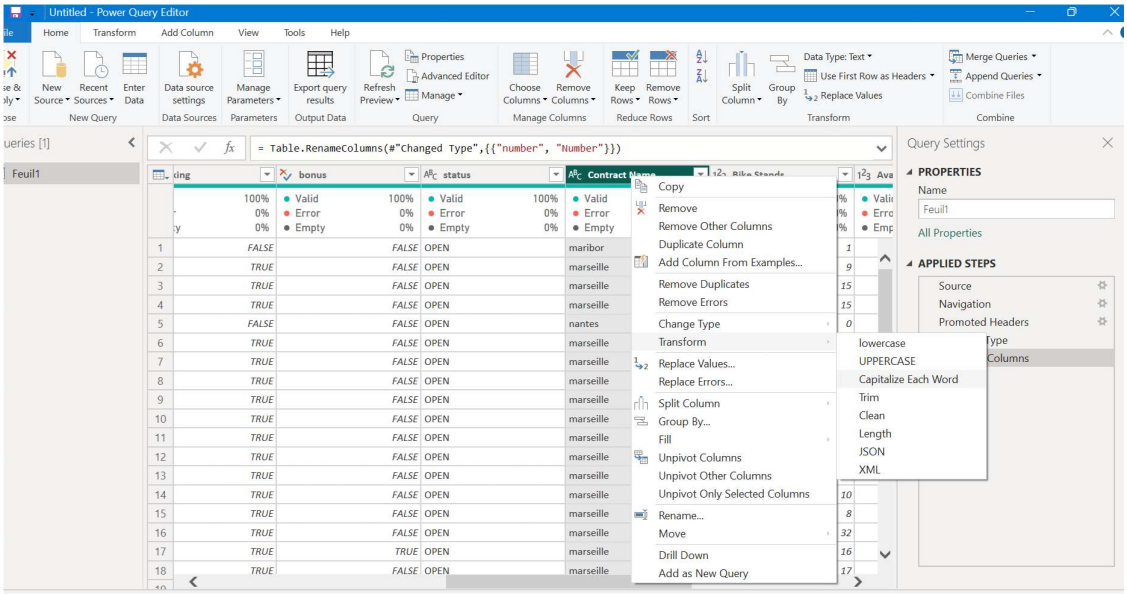
- **Import file:**



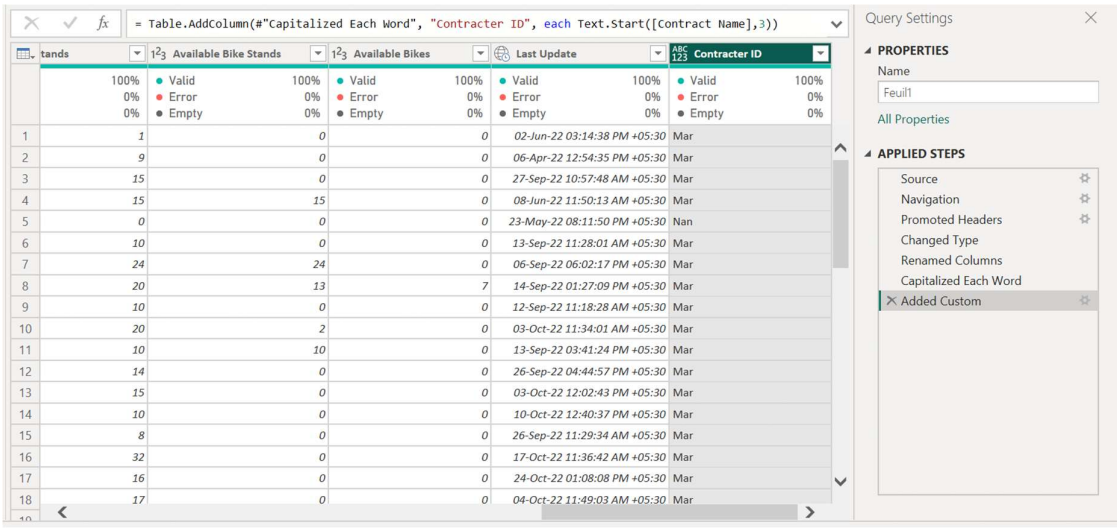


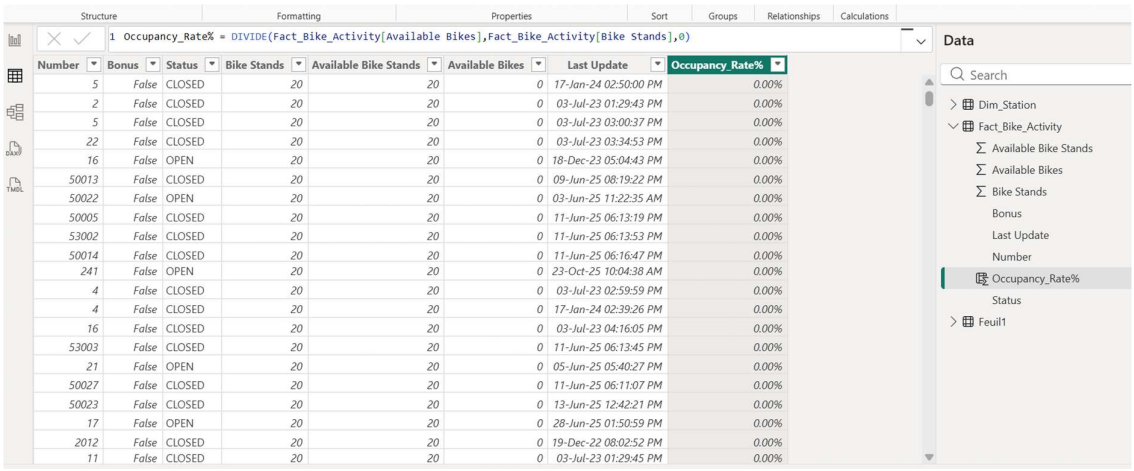
number	name	address	position
2000	City Central Bike Station 1	102 Main Street, City	46.5488766782
1297	City Central Bike Station 2	103 Main Street, City	43.2934857361
8265	City Central Bike Station 3	104 Main Street, City	43.2693185563
8065	City Central Bike Station 4	105 Main Street, City	43.2672725156
1011	City Central Bike Station 5	106 Main Street, City	47.19504, -1.55
4342	City Central Bike Station 6	107 Main Street, City	43.3059934537
6176	City Central Bike Station 7	108 Main Street, City	43.2847759836
3322	City Central Bike Station 8	109 Main Street, City	43.3123370339
1264	City Central Bike Station 9	110 Main Street, City	43.2978963132
1191	City Central Bike Station 10	111 Main Street, City	43.2962076706
5058	City Central Bike Station 11	112 Main Street, City	43.2928008635
4293	City Central Bike Station 12	113 Main Street, City	43.2962634051
1245	City Central Bike Station 13	114 Main Street, City	43.2990279382
7153	City Central Bike Station 14	115 Main Street, City	43.2894676088
6154	City Central Bike Station 15	116 Main Street, City	43.2903297702
1002	City Central Bike Station 16	117 Main Street, City	43.2961387765
1224	City Central Bike Station 17	118 Main Street, City	43.2958251832
1287	City Central Bike Station 18	119 Main Street, City	43.2995209755
500	City Central Bike Station 19	120 Main Street, City	48.799828, 1.9
5220	City Central Bike Station 20	121 Main Street, City	43.2989919565
6160	City Central Bike Station 21	122 Main Street, City	43.2868383512
7026	City Central Bike Station 22	123 Main Street, City	43.2898562014

- **Data Cleaning & Transformation:** Removed duplicates, handled missing values, standardized formats, and created calculated fields.

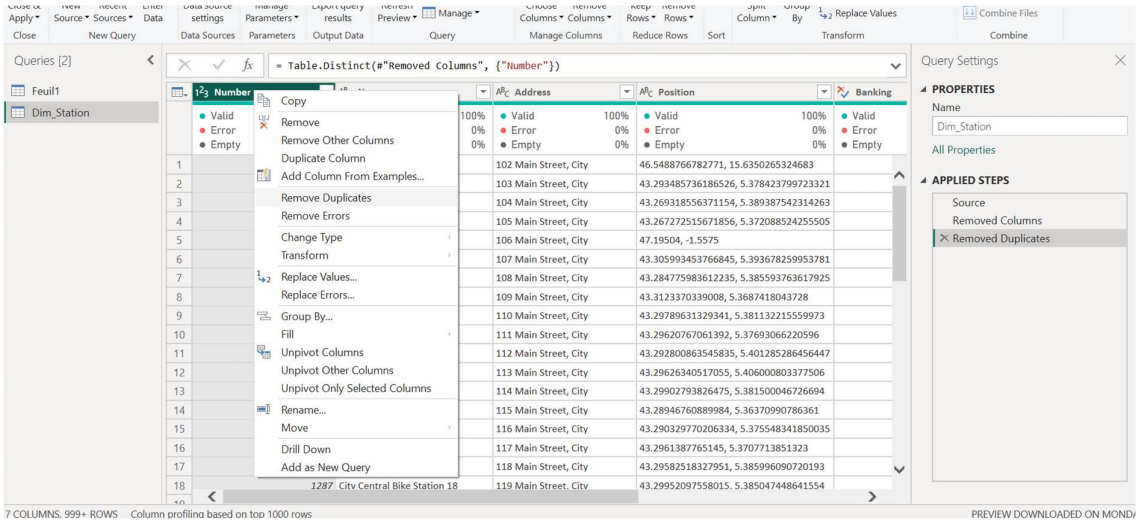


- **Created column**

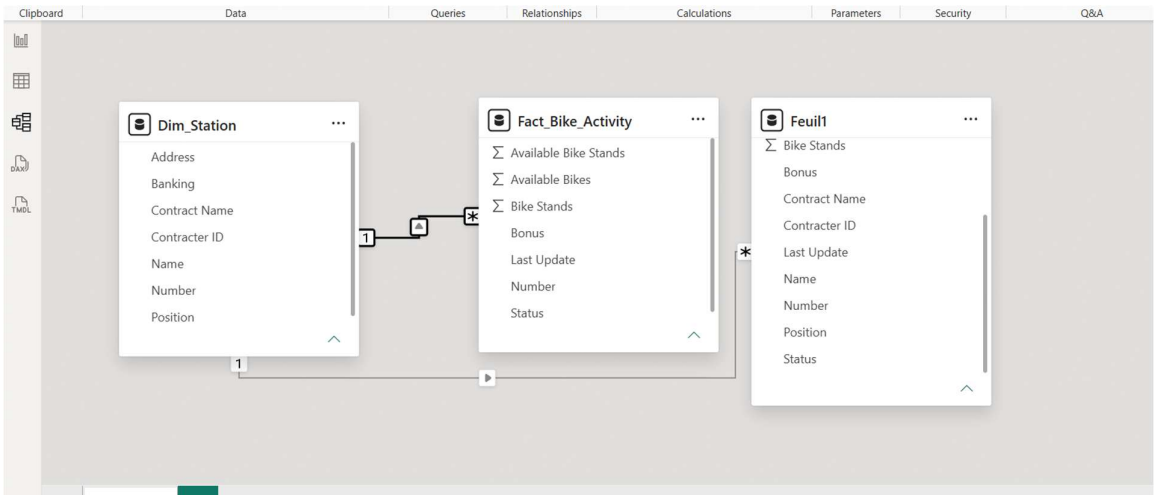




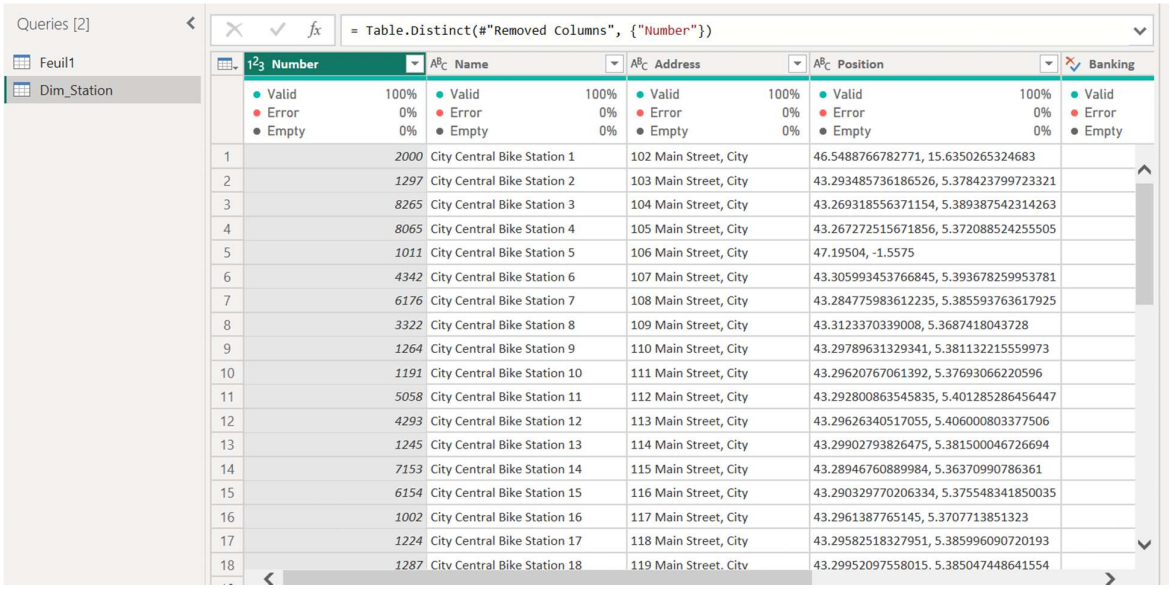
Removed Duplicates:



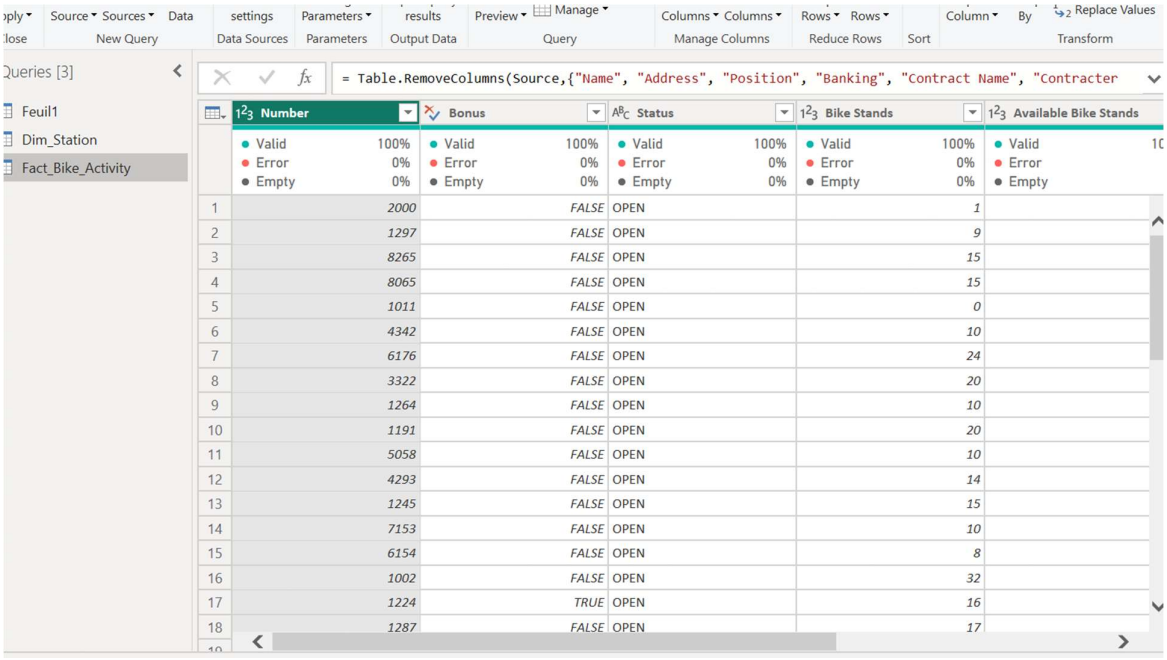
Convert the data into Fact and Dimension Table



FACT AND DIM TABLE CREATED:



	Number	Name	Address	Position	Banking
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid Error Empty
1	2000	City Central Bike Station 1	102 Main Street, City	46.5488766782771, 15.6350265324683	
2	1297	City Central Bike Station 2	103 Main Street, City	43.293485736186526, 5.378423799723321	
3	8265	City Central Bike Station 3	104 Main Street, City	43.269318556371154, 5.389387542314263	
4	8065	City Central Bike Station 4	105 Main Street, City	43.267272515671856, 5.372088524255505	
5	1011	City Central Bike Station 5	106 Main Street, City	47.19504, -1.5575	
6	4342	City Central Bike Station 6	107 Main Street, City	43.305993453766845, 5.393678259953781	
7	6176	City Central Bike Station 7	108 Main Street, City	43.284775983612235, 5.385593763617925	
8	3322	City Central Bike Station 8	109 Main Street, City	43.3123370339008, 5.3687418043728	
9	1264	City Central Bike Station 9	110 Main Street, City	43.29789631329341, 5.381132215559973	
10	1191	City Central Bike Station 10	111 Main Street, City	43.29620767061392, 5.37693066220596	
11	5058	City Central Bike Station 11	112 Main Street, City	43.292800863545835, 5.401285286456447	
12	4293	City Central Bike Station 12	113 Main Street, City	43.29626340517055, 5.406000803377506	
13	1245	City Central Bike Station 13	114 Main Street, City	43.29902793826475, 5.381500046726694	
14	7153	City Central Bike Station 14	115 Main Street, City	43.28946760889984, 5.36370990786361	
15	6154	City Central Bike Station 15	116 Main Street, City	43.290329770206334, 5.375548341850035	
16	1002	City Central Bike Station 16	117 Main Street, City	43.2961387765145, 5.3707713851323	
17	1224	City Central Bike Station 17	118 Main Street, City	43.29582518327951, 5.385996090720193	
18	1287	City Central Bike Station 18	119 Main Street, City	43.29952097558015, 5.385047448641554	

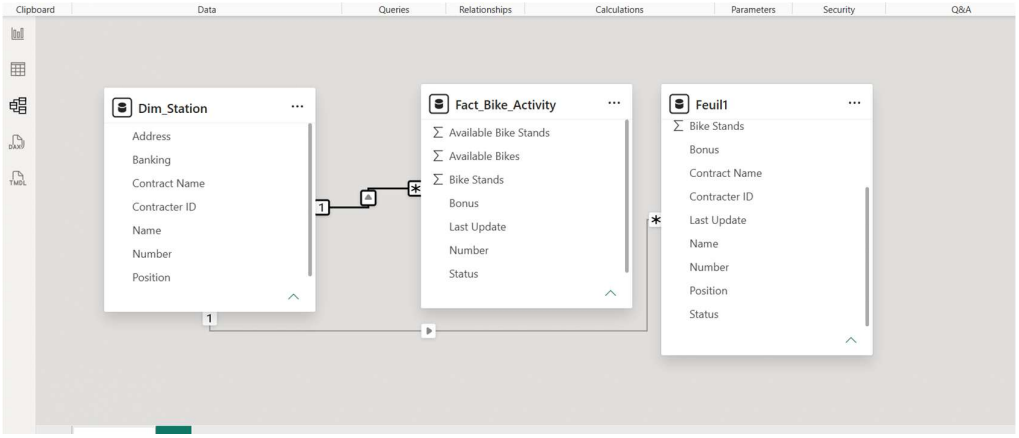


	Number	Bonus	Status	Bike Stands	Available Bike Stands
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid Error Empty
1	2000	FALSE	OPEN	1	
2	1297	FALSE	OPEN	9	
3	8265	FALSE	OPEN	15	
4	8065	FALSE	OPEN	15	
5	1011	FALSE	OPEN	0	
6	4342	FALSE	OPEN	10	
7	6176	FALSE	OPEN	24	
8	3322	FALSE	OPEN	20	
9	1264	FALSE	OPEN	10	
10	1191	FALSE	OPEN	20	
11	5058	FALSE	OPEN	10	
12	4293	FALSE	OPEN	14	
13	1245	FALSE	OPEN	15	
14	7153	FALSE	OPEN	10	
15	6154	FALSE	OPEN	8	
16	1002	FALSE	OPEN	32	
17	1224	TRUE	OPEN	16	
18	1287	FALSE	OPEN	17	

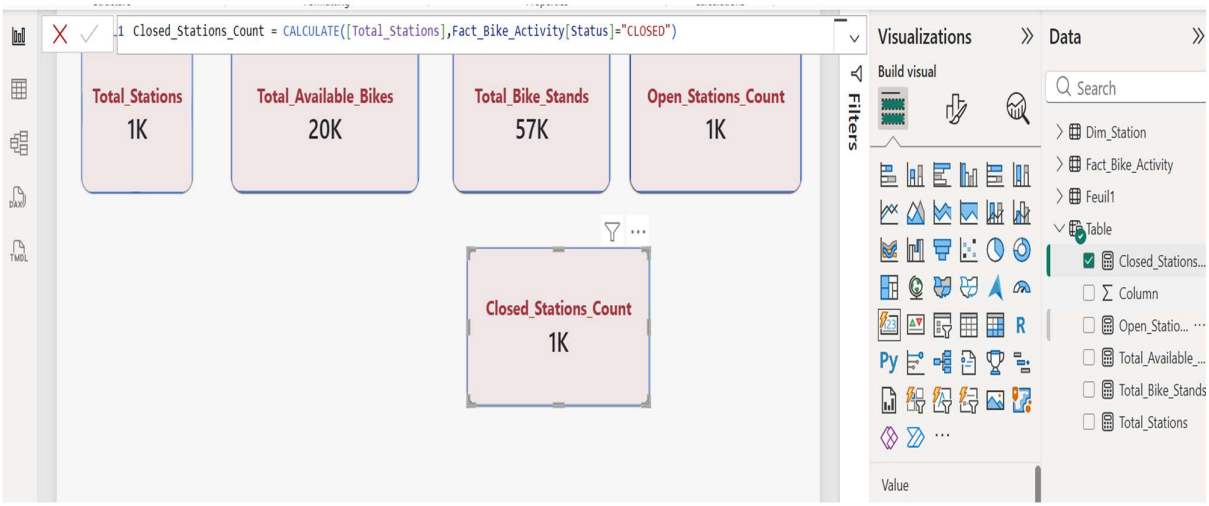
Data Modelling and DAX (Power BI)

- **Data Model:** Established relationships between tables, defined cardinality, and created lookup tables where necessary.

SCREENSHOT OF DATA MODELLING



- **Calculated Columns & DAX Measures:** Implemented DAX formulas for key metrics, such as total sales, growth percentages, and cumulative totals**.**




```
Closed_Stations_Count = CALCULATE([Total_Stations],Fact_Bike_Activity[Status]="CLOSED")

Open_Stations_Count = CALCULATE([Total_Stations],Fact_Bike_Activity[Status]="OPEN")

Total_Available_Bikes = SUM(Fact_Bike_Activity[Available Bikes])

Total_Bike_Stands = SUM(Fact_Bike_Activity[Bike Stands])

Total_Stations = DISTINCTCOUNT(Dim_Station[Number])
```

Analysis and Visualizations (Power BI)

Dashboard Features:

- **Multiple Visualizations based on problem statement:**

Status Distribution - Open Vs Closed (Pie / Donut Chart)

Description: Represents the proportion of active versus inactive bike stations across the network.

Key Insight: Shows an equal 50%-50% distribution (1K Open vs. 1K Closed), highlighting half the network capacity in each operational state.

2. Contract Analysis by City / Network (Clustered Column Chart)

Description: Compares average bike stand capacities across different city contracts and municipal networks.

Key Insight: Displays comparative benchmarks for major contracts such as Dublin, Bruxelles-Capitale, Lyon, and Toulouse.

3. Banking Availability (Horizontal Bar Chart)

Description: Evaluates payment infrastructure availability (Banking = True / False) across individual contracts and cities.

Key Insight: Helps identify which municipal networks support on-site credit/debit card transactions.

4. Last Update & Occupancy Rate Trend (Area / Line Chart)

Description: Tracks historical occupancy rate percentages across a multi-year timeline (spanning July 2022 to July 2025).

Key Insight: Uncovers cyclical demand fluctuations and long-term usage trends over time.

5. Top Contracts by Bike Stands (Horizontal Bar Chart)

Description: Ranks municipal contracts based on total sum of available bike stand capacity.

Key Insight: Highlights top-tier networks like Lyon, Toulouse, and Bruxelles-Capitale as primary infrastructure hubs.

6. Bike Station Map & Status Overview (Geographic Map Visual)

Description: Displays the spatial distribution of bike stations mapped geographically across European regions.

Key Insight: Uses color-coded indicators (Open vs. Closed) to provide instant geospatial operational visibility.

7. Contract Status & Bike Availability Summary (Matrix / Table Visual)

Description: Provides detailed tabular summaries broken down by Contract Name, Status, Sum of Bike Stands, and Sum of Available Bikes.

Key Insight: Offers precise numerical validation for contract-level inventory auditing.

8. Bike Stand Vs Available Bike Analysis (Scatter Plot)

Description: Plots total stand capacity against currently available bikes for each station.

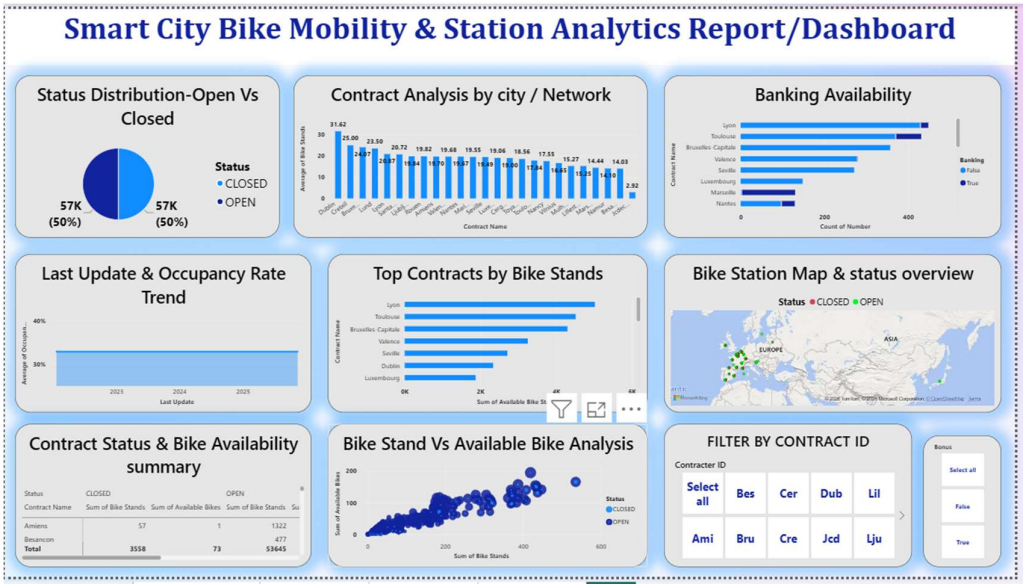
Key Insight: Identifies correlation patterns and detects outliers where station capacity is underutilized or over utilized.

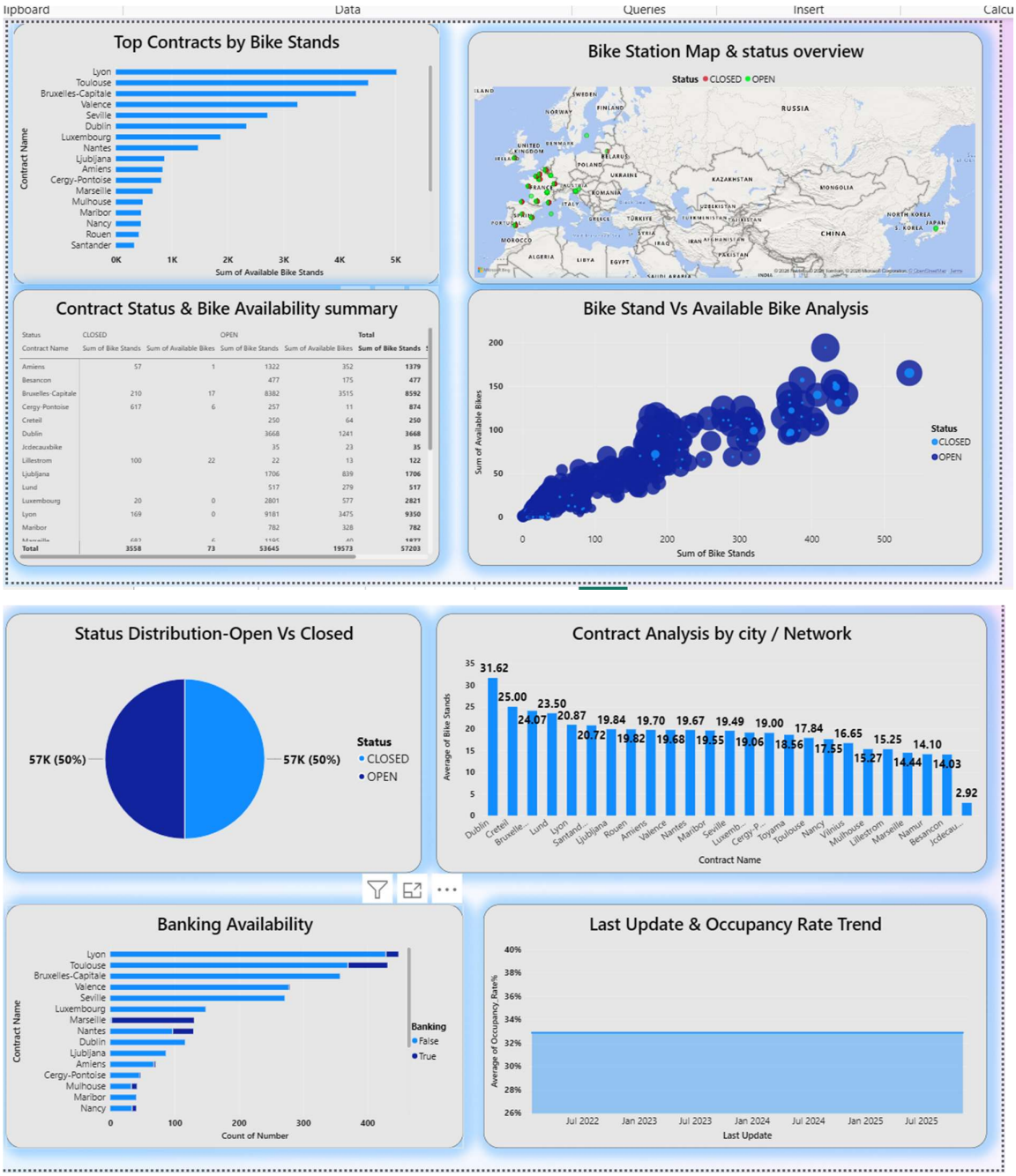
- **Make your Report interactive Layout using Filters and slicers:**

Filter by Contract ID (Slicer / Button Grid): Allows users to dynamically slice report data by specific contractor identifiers (e.g., Ami, Bes, Cer, Dub, Lyo).

- **Create a consolidated Report /Dashboard.**

SCREENSHOT OF DASHBOARD /REPORT




lipboard
Data
Queries
Insert
Calcu
Top Contracts by Bike Stands
Contract Name
Lyon
Toulouse
Bruxelles-Capitale
Valence
Seville
Dublin
Luxembourg
Nantes
Ljubljana
Amiens
Cergy-Pontoise
Marseille
Mulhouse
Maribor
Nancy
Rouen
Santander
0K
1K
2K
3K
4K
5K
Sum of Available Bike Stands
Bike Station Map & status overview
Status CLOSED OPEN
Contract Status & Bike Availability summary
Status
CLOSED
OPEN
Total
Contract Name
Sum of Bike Stands
Sum of Available Bikes
Sum of Bike Stands
Sum of Available Bikes
Sum of Bike Stands
Amiens 57 1 1322 352 1379
Besancon 477 175 477
Bruxelles-Capitale 210 17 8382 3515 8592
Cergy-Pontoise 617 6 257 11 874
Creteil 250 64 250
Dublin 3668 1241 3668
Jcdecauxbike 35 23 35
Lillestrom 100 22 22 13 122
Ljubljana 1706 839 1706
Lund 517 279 517
Luxembourg 20 0 2801 577 2821
Lyon 169 0 9181 3475 9350
Maribor 782 328 782
Total 3558 73 53645 19573 57203
Bike Stand Vs Available Bike Analysis
Sum of Available Bikes
Sum of Bike Stands
Status
CLOSED
OPEN
Status Distribution-Open Vs Closed
57K (50%)
57K (50%)
Status
CLOSED
OPEN
Contract Analysis by city / Network
Average of Bike Stands
31.62
25.00
24.07
23.50
20.87
20.72
19.84
19.82
19.70
19.68
19.67
19.55
19.49
19.06
19.00
18.56
17.84
17.55
16.65
15.27
15.25
14.44
14.10
14.03
2.92
Dublin
Creteil
Bruxelle...
Lund
Lyon
Santand...
Ljubljana
Rouen
Amiens
Valence
Nantes
Maribor
Seville
Luxemb...
Cergy-P...
Toyama
Toulouse
Nancy
Vilnius
Mulhouse
Lillestrom
Marseille
Namur
Besancon
Jcdecau...
Contract Name
Banking Availability
Contract Name
Lyon
Toulouse
Bruxelles-Capitale
Valence
Seville
Luxembourg
Marseille
Nantes
Dublin
Ljubljana
Amiens
Cergy-Pontoise
Mulhouse
Maribor
Nancy
Banking
False
True
0
100
200
300
400
Count of Number
Last Update & Occupancy Rate Trend
Average of Occupancy_Rate%
40%
38%
36%
34%
32%
30%
28%
26%
Jul 2022
Jan 2023
Jul 2023
Jan 2024
Jul 2024
Jan 2025
Jul 2025
Last Update

Insights & Conclusions

Provide the analysis insights:

- **Descriptive**

  Total Stations: Approximately 1K (1,028 stations).

  Operational Status: The ratio of open to closed stations is evenly split at 50% - 50% (Open: 1K, Closed: 1K).

  Total Bike Stands: Recorded as 57K, with Total Available Bikes at 20K.

  Major cities/contracts like Lyon, Toulouse, and Bruxelles-Capitale account for a higher number of bike stands and service coverage.

- **Diagnosis**

  Based on the Contract Analysis by city / Network and Banking Availability charts, specific contracts (such as Lyon and Toulouse) exhibit higher banking availability and bike stand utilization.

  According to the Fact_Bike_Activity table, even though certain stations are marked as CLOSED, stand allocations and bike provisions continue to be maintained. Operational quality varies based on occupancy rates and last-update timelines.

- **Predictive**

  Through the Last Update & Occupancy Rate Trend and scatter plot (Bike Stand Vs Available Bike Analysis) visuals, the occupancy rate has experienced cyclical fluctuations over the past few years (from 2022 to 2025).

  By analyzing the relationship between the number of stands and available bikes, models can help predict future surges in bike demand across various cities.

- **Prescriptive**

  Since 50% of the stations are currently closed, investigate the reasons for low demand in those regions and consider reopening them or reallocating bikes to high-demand areas like Lyon and Toulouse.

  Utilize the map overview (Bike Station Map & status overview) to pinpoint underserved locations and optimize urban infrastructure planning.

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10. Conclusions

This Smart City Bike Mobility assignment successfully analyzes network-wide bike stand utilization, open versus closed station balance, and occupancy rate trends. These insights can be leveraged to further enhance urban transportation management.